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A Biomimetic Multimodal Whisker Sensor for Both Contact and Airflow Sensing of Mobile Robots

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ABSTRACT

Multimodal sensing is critical for robust environmental perception, especially for mobile robots operating in unstructured or visually degraded environments. This paper introduces a compact, biomimetic whisker sensor capable of both contact localization along the whisker shaft and airflow characterization. The sensor employs an elastic four-bar mechanism instrumented with a small set of strain gauges to capture local deformations. A kinetostatics model of the whisker is first established, and a cascaded residual learning framework that combines the physics-based model with a neural network is developed to estimate the spatial coordinates of contact points. For airflow sensing, a simplified model is introduced to guide the signal processing, enabling the identification of wind direction and estimation of flow speed from strain gauge responses. Experimental results validate the sensor's accuracy in both modalities. Furthermore, multiple identical sensors are integrated on a mobile robot, achieving anemotactic navigation and obstacle avoidance. These results highlight the sensor's potential as a lightweight, low-cost, and multifunctional solution for multimodal environmental perception on robotic platforms.

1. Introduction

In nature, many animals, especially mammals, use whiskers to perceive and interact with their surroundings [1, 2, 3]. These whiskers possess multimodal sensory capabilities. Taking rats as an example, they can not only detect direct contact with external obstacles but also identify the source of wind by sensing airflow [4, 5]. Inspired by these biological systems, bionic whisker sensors have been extensively developed and applied in robotic systems, substantially enhancing environmental perception [6, 7]. Existing whisker sensors largely fall into two categories: those designed for contact detection, and those developed for fluid flow sensing in air or water. While these sensors have achieved notable success in specific modalities, they remain limited in their ability to simultaneously perceive both contact and flow, which is a hallmark of biological whiskers.

Most whisker sensors have been developed to detect direct physical contact with objects. These sensors employ single or multiple sensing elements based on various working principles, including piezoelectric, capacitive, optical and resistive mechanisms, achieving contact detection through either passive or active sensing modalities [8, 9, 10, 11].

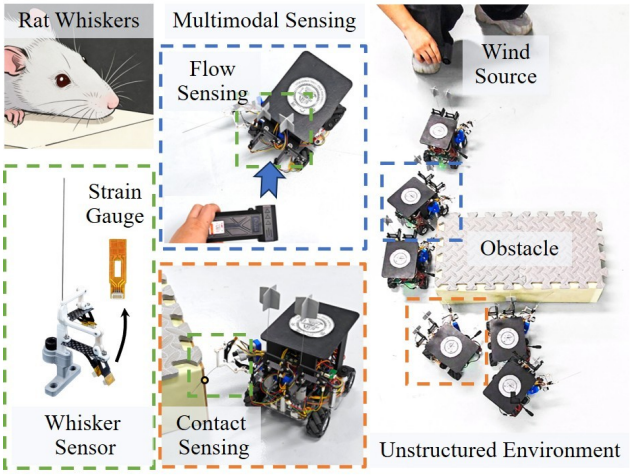


Figure 1: Whisker Sensor for Multimodal Sensing in Unstructured Environment.

Pegoli *et al.* optimize electromechanical whisker design using beam theory and a linear approximation of strain rate, achieving accurate contact localization with an optimally tapered whisker[12]. Yu *et al.* developed an active whisker sensor for contact sensing of different characteristics including distance, shape, size, and orientation. Application is also demonstrated on mobile robots by sweeping the hairy whiskers actively to interact with the environment [13] .

Another category of whisker sensors has been developed for the detection of fluids such as air or water [14, 15].

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Kim *et al.* developed a whisker-inspired sensor based on magnetic transduction for multi-directional airflow sensing, which achieves passive measurement of both airflow speed and direction [16]. Qi *et al.* developed a self-powered artificial vibrissal system that can synchronously perceive and encode wind speed, direction, humidity, and temperature [17]. Liu *et al.* present a bionic whisker sensor that mimics the undulating morphology of harbor seal whiskers, which is able to detect oscillatory flows and identify upstream vortex wakes, showcasing its capability for detailed underwater flow analysis [18].

Recent advances in functional materials and energy-efficient sensing technologies have further expanded the capabilities of sensor systems. For instance, triboelectric nanogenerator (TENG)-based sensors have demonstrated promising potential for self-powered environmental monitoring and robotic perception[19, 20, 21, 22, 23]. Neuromorphic devices have also been developed for intelligent visual information processing[24]. These developments highlight the growing interest in creating energy-autonomous, material-innovative, and versatile sensing solutions.

While existing whisker sensors have achieved a certain level of environmental detection, they still exhibit certain limitations. On one hand, current whisker sensors for direct contact perception are often limited to qualitative detection and face significant challenges in accurately determining contact points when they are not located at the whisker tip. Despite the achievement of intermediate contact sensing of several studies [25, 26, 27], current implementations necessitate bulky force/torque measurement and sampling devices, compromising the system's compactness. On the other hand, most existing whisker sensors are generally designed to perceive only a single environmental characteristic of direct contact or flow. Several existing studies have developed multimodal whisker sensors [28, 29, 30], however, they still face notable limitations. Primarily, these systems often rely on complex hardware architectures involving multiple embedded sensors or intricate visual sensing units, leading to high fabrication costs and integration challenges. Furthermore, many prototypes have not been robustly integrated into functional robotic systems, leaving their real-world performance and reliability unverified.

To address these challenges, there is a pressing need for whisker sensors that are structurally simple, low-cost, and capable of integrating multiple sensing modalities to enable comprehensive environmental perception in robotic systems. This paper proposes a bio-inspired whisker sensor for multimodal sensing of direct contact and airflow. As shown in Fig. 1, based on the continuum mechanics of slender elastic links, which has been proved to be compact, reliable and adaptable to environment [31, 32], the proposed whisker sensor features a simple structure with low fabrication cost. Industrial-grade strain gauges are used for local deformation detection, enabling the identification of spatial contact points at intermediate location along the whisker shaft through a proposed model-based approach. Furthermore, through analysis of strain gauge responses under airflow, the sensor can determine wind direction and estimate flow speed, demonstrating multimodal sensing within a unified hardware platform. Compared to existing tactile and whisker-inspired sensors such as BioTacTip[33], TacTip[34], BIOTACT[35], and WhiskSight[28], the proposed sensor achieves both contact localization and airflow characterization within a single, mechanically simple hardware platform using only four strain gauges. It requires no cameras or complex optical components, enables real-time processing on low-power microcontrollers, and offers inherent mechanical compliance.

In addition to experimental validation, the developed sensors were also applied to a robotic system for demonstration. Recent studies have shown that rats achieve multimodal perception through functional specialization within their whisker array: the supra-orbital whiskers act as dedicated airflow sensors, while the mystacial whiskers primarily mediate tactile exploration [36]. Inspired by this biological division of labor, we adopt multiple whisker sensors of identical mechanical and instrumental design on a mobile robot, which can be configured for either modality. This hardware uniformity enables multimodal environmental perception without requiring sensor redesign, thereby reducing fabrication complexity and cost. Demonstrations are provided to illustrate airflow-guided navigation and obstacle avoidance in unstructured environments.

The main contributions of this work are as follows:

- A bio-inspired whisker sensor design that integrates both contact and airflow sensing modalities using strain-based deformation measurement.
- Spatial contact point localization and flow sensing methodology, supported by model-based analysis, experimental calibration and testing.
- On-robot integration and demonstrations showing anemotaxis and obstacle avoidance in unstructured environments.

The rest of this paper is structured as follows: Section 2 presents the design and prototype of the proposed sensor. Section 3 establishes the kinetostatics model of the whisker sensor. Section 4 describes the multimodal sensing methods, including direct contact localization and airflow characterization. Section 5 demonstrates field applications by implementing the sensor on a mobile robot for anemotaxis and obstacle avoidance. Section 6 discusses the assumptions, advantages, and limitations of the approach. Finally, Section 7 concludes the paper and outlines future research directions.

2. Sensor Design and Prototype

Inspired by biological whiskers, the proposed whisker sensor adopts a similar structure. As depicted in Fig. 2, the sensor consists of a whisker shaft with sensing elements concentrated at its base, analogous to the follicular structure of biological whiskers. The whisker shaft is fabricated from a 0.6 mm diameter nitinol alloy rod, with a length of 80

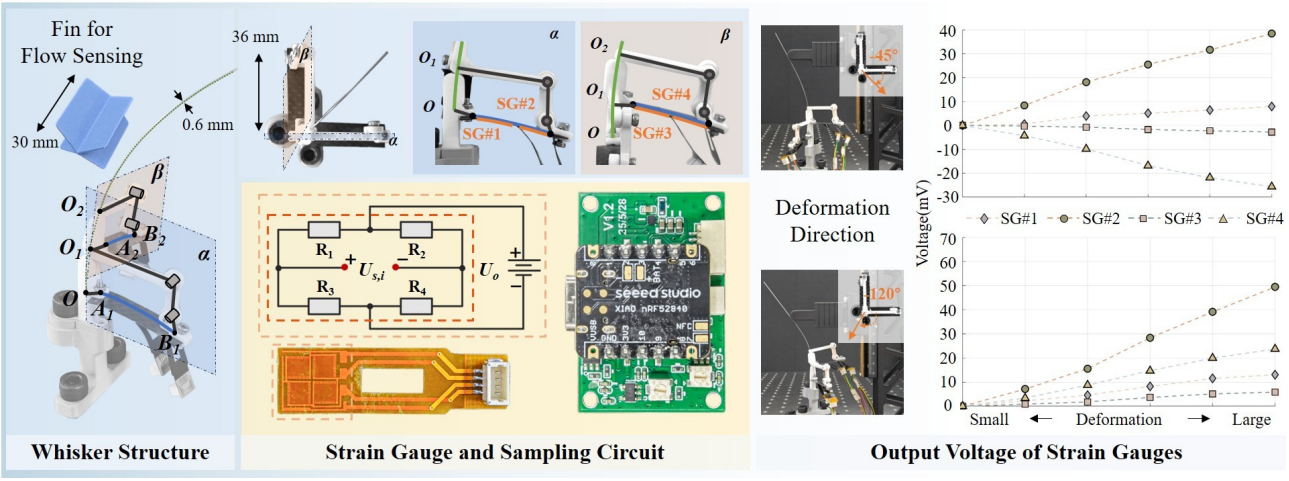


Figure 2: Structure and the Measuring System for the Developed Whisker Sensors.

mm from O_2 to the free end. The sensing module comprises two four-bar linkage mechanisms connected in series and oriented in two perpendicular planes α and β.

To constrain the motion of the two four-bar linkages within their respective planes, baffle plates are incorporated on the sides of the whisker shaft. This design ensures that segment OO_1 of the whisker shaft bends exclusively within plane α, while segment O_1O_2 bends exclusively within plane β. During the motion of the mechanism, the normal vector of plane α remains unchanged, whereas the normal vector of plane β varies in conjunction with the bending of the whisker shaft and remains perpendicular to the tangent direction of the shaft at point O_1.

Within each four-bar linkage, one specific link (A_1B_1 and A_2B_2 respectively) is constructed using a 0.2 mm thin and 36 mm long elastic carbon fiber beam, with two strain gauges (SG) deployed to its surface. When an external force is applied to the whisker shaft, it undergoes deformation, which subsequently causes displacement in the two four-bar linkage mechanisms. This results in the bending of the elastic beams and the attached strain gauges, which can thus be measured and reflects the characteristics of the applied external force. The remaining rigid links and the whisker base were fabricated via 3D printing using polylactic acid (PLA) material, enabling low-cost and rapid whisker fabrication.

Due to the slender nature of the whisker shaft, airflow alone cannot generate sufficient thrust. Therefore, when deployed for airflow detection, a PLA-printed fin-like structure is installed at the free distal end of the whisker. This addition increases the contact surface area exposed to the airflow, thereby enabling the elastic beam to undergo significant deformation.

For the developed sensor, specialized strain gauges were fabricated. As shown in Fig. 2, they employ a full-bridge configuration to reduce noise and are directly manufactured on flexible printed circuit boards, eliminating the need for additional soldering. This design offers high integration and stability. Each full-bridge strain gauge produces a single differential output voltage, which is proportional to the local bending angle. The current sensor design incorporates four strain gauges, providing four independent output channels. A sampling circuit incorporating a 24-bit analog to digital converter AD7124 was developed to acquire the output voltages from the strain gauges, and a compact Seeed Studio XIAO nRF52840 was chosen as the microcontroller. This circuit enables simultaneous sampling of 4 strain gauge channels at a frequency exceeding 100 Hz, while maintaining a compact size and low power consumption. The entire sensor system operates on a single 3.7V lithium battery and is capable of transmitting the acquired data wirelessly.

To demonstrate the performance of the whisker sensor prototype and the associated sampling system, the whisker shaft was deflected in multiple distinct directions through direct contact, with three repeated trials conducted for each direction. Fig. 2 illustrates the variations in output voltage from the four strain gauges as for two of these directions. Under identical contact conditions, the maximum fluctuation in output voltage was maintained below 2 mV. As the figure shows, when the whisker shaft is bent in different directions, the four strain gauges exhibit distinct voltage response patterns, validating the sensor's capability for directional differentiation. Consequently, the deformation state of the whisker shaft—and thus the nature of the stimulus, including direct contact and fluid flow—can be deduced by analyzing the voltage values from the strain gauges. The voltage-deformation curves presented here do not imply a relationship applied to all configurations, a global sensing model is required to capture the mapping between the strain gauge measurements and the external contact conditions. Moreover, the developed sensor and sampling system exhibit good reliability and can operate stably even under repeated and long-term loading. The results of repeated loading and fatigue tests are shown in the Supplementary Materials.

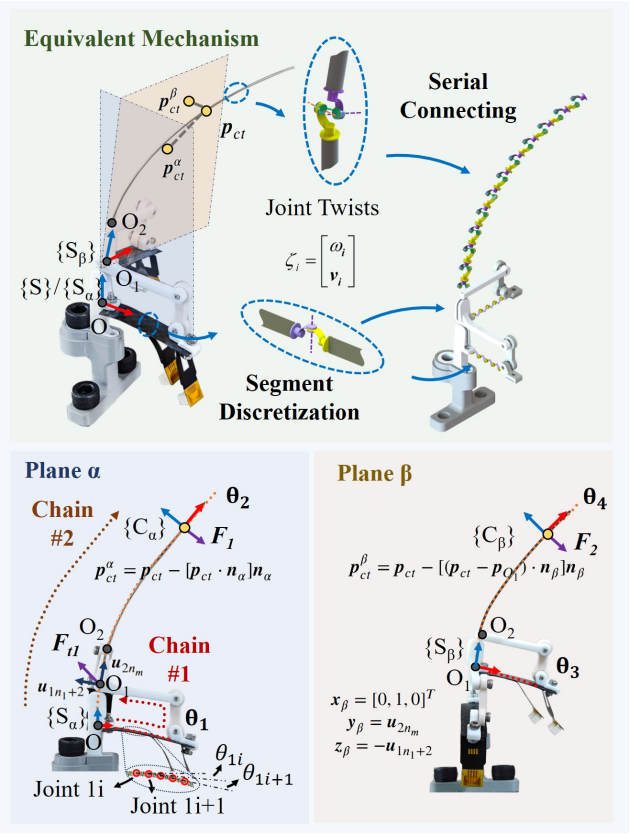


Figure 3: Equivalent Mechanism and Kinetostatics Modeling Process of the Proposed Sensors.

3. Kinetostatics Modeling of the Whisker Sensor

Based on our prior work [37], elastic links can be discretized into a number of small segments. For each segment, a spatial six-DOF linkage with rigid bodies and passive elastic joints can be obtained [38], which can approximate the force deflection behavior of the segments. By connecting all the segments, a hyper-redundant rigid-body mechanism with elastic joints can be constructed. For the proposed sensor, the effects of shearing and compression are neglectable comparing to the bending one. As a result, only the bending related revolute joints are considered in the following modeling process. Specifically, for the elastic beams, only single-DOF bending is considered, while for the whisker shaft, bending with two DOF is taken into account. In earlier research, we have established and validated a kinetostatics model of a single elastic beam using the discretizing approach [39]. Here, we extend this model to the more complicated whisker sensor.

As shown in Fig. 3, to facilitate characterization, a spatial coordinate system is first established for the whisker sensor. In the undeformed state of the whisker, the intersection point of the centerline of the lower elastic beam and the whisker shaft is taken as the origin. The centerline of the lower elastic beam serves as the x-axis, with the positive direction pointing outward. The direction of the centerline of the upper elastic beam is defined as the y-axis, with the positive direction pointing outward. The z-axis coincides with the whisker shaft, and its positive direction points toward the free end.

As described before, although the developed whisker sensor bends in space, the motion of the two elastic beams is confined to a plane due to the presence of baffle plates. Therefore, during modeling, the spatial motion of the whisker can be represented by the superposition of two planar motions, with the motions of the two planes forming a serial relationship. First, consider the motion in the α plane, the planar mechanism can be divided into two chains for modeling. The first chain includes the elastic beam and the four-bar mechanism, while the second chain comprises the whisker shaft. Using product-of-exponential (POE) formulation, for the first kinematic chain, the end coordinate frame of the equivalent linkage mechanism can be expressed as:

$$\mathbf{g}_{st1}(\boldsymbol{\theta_1}) = \exp(\hat{\boldsymbol{\xi}}_{11}\theta_{11}) \cdots \exp(\hat{\boldsymbol{\xi}}_{1n_1+2}\theta_{1n_1+2})\,\mathbf{g}_{st1,0} \quad (1)$$

where $\hat{\boldsymbol{\xi}}_{1i} \in se(3)$, $i = 1, \cdots, n+2$ denote the twists of the joints in home configuration. $\boldsymbol{\theta}_1 = \left[\theta_{11}, \cdots, \theta_{1n_1+2}\right] \in \mathbb{R}^{(n_1+2)\times 1}$ denotes the bending related joint displacement of the whole link. n_1 is the number of equivalent joints for the elastic beam. Since the first kinematic chain includes two rigid links of the four-bar mechanism, its end incorporates two passive revolute joints. Thus, the total number of joints in the first kinematic chain is $n_1 + 2$. Correspondingly, the joint force balance constraint for the first kinematic chain can be expressed as:

$$\boldsymbol{\tau_1} = \mathbf{K}_{\boldsymbol{\theta_1}}\,\boldsymbol{\theta_1} - \mathbf{J}_{t_1}^T\,\boldsymbol{F}_{t_1} = \boldsymbol{0} \quad (2)$$

$\mathbf{J}_{\mathbf{t_1}} = \left[\boldsymbol{\xi}_{11}, \cdots, \boldsymbol{\xi}_{1n_1+2}\right] \in \mathbb{R}^{6\times(n_1+2)}$ is the Jacobian matrix. Note that in the diagonal stiffness matrix $\mathbf{K}_{\boldsymbol{\theta_1}} = \mathrm{diag}(k_{11}, \cdots, k_{1n_1+2}) \in \mathbb{R}^{n\times n}$, the stiffness of the last two joints (passive revolute joints) should be zero.

For the developed whisker sensor, considering frictionless single-point contact on the whisker shaft. A spatial frame $\{\mathrm{S}_\alpha\}$ and a floating contact frame $\{\mathrm{C}_\alpha\}$ are constructed at the base and the contact point of the beam. The following geometric constraint must be satisfied for the second kinematic chain:

$$\mathbf{g}_{sc1}(\boldsymbol{\theta_2}) = \exp(\hat{\boldsymbol{\xi}}_{21}\theta_{21}) \cdots \exp(\hat{\boldsymbol{\xi}}_{2n_{c1}}\theta_{2n_{c1}})\,\mathbf{g}_{sc1,0} = \mathbf{g}_{c1} \quad (3)$$

where $\mathbf{g}_{sc1}$ represents the relative pose of the contact frame $\{\mathrm{C}_\alpha\}$ with respect to $\{\mathrm{S}_\alpha\}$, and n_{c1} denotes the number of joints before the contact position. $\mathbf{g}_{c1}$ is the target contact pose introduced by the geometric constraints of the contact object. Additionally, the joint force balance constraint must be satisfied:

$$\boldsymbol{\tau_2} = \mathbf{K}_{\boldsymbol{\theta_2}}\,\boldsymbol{\theta_2} - \mathbf{J}_1^T\,\boldsymbol{F}_1 + \mathbf{J}_{m2}^T\,\boldsymbol{F}_{t_1} = \boldsymbol{0} \quad (4)$$

Note that $\boldsymbol{F}_{t_1}$ in $\boldsymbol{\tau}_1$ represents the force exerted by the whisker shaft on the four-bar mechanism. Consequently,

the reaction force from the four-bar mechanism on the whisker shaft is $-\boldsymbol{F}_{t_1}$, resulting in the term $\mathbf{J}_{m2}\boldsymbol{F}_{t_1}\cdot\mathbf{J}_m = \left[\boldsymbol{\xi}_{21},\cdots,\boldsymbol{\xi}_{2n_m},\boldsymbol{0},\cdots,\boldsymbol{0}\right] \in \mathbb{R}^{6\times n_2}$ is related to the number of joints n_m at the connection between the first and second kinematic chains, and n_2 is the total number of segments in the whisker shaft's elastic linkage. Furthermore, since the end of the first kinematic chain is rigidly connected to the second and always remains perpendicular, the local coordinate frame $\mathbf{g}_{sm}$ at the end of the first kinematic chain must satisfy the following geometric relationship:

$$\mathbf{g}_{sm2}(\boldsymbol{\theta_2}) = \exp(\hat{\boldsymbol{\xi}}_{21}\theta_{21})\cdots\exp(\hat{\boldsymbol{\xi}}_{2n_m}\theta_{2n_m})\,\mathbf{g}_{sm2,0} = \mathbf{g}_{st1}\mathbf{T}_{12} \tag{5}$$

where $\mathbf{T}_{12}$ is the coordinate transformation matrix representing the perpendicular relationship. Based on the above analysis process, the kinetostatics model of the whisker's motion in plane α can be expressed in the following form:

$$\boldsymbol{C}_\alpha(\boldsymbol{x}_\alpha) = \begin{bmatrix} \boldsymbol{\tau_1} \\ \boldsymbol{\tau_2} \\ \boldsymbol{y}_{c1} \\ \boldsymbol{y}_{t1} \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\theta_1}\boldsymbol{\theta_1} - \mathbf{J}_{t_1}^T\boldsymbol{F}_{t_1} \\ \mathbf{K}_{\theta_2}\boldsymbol{\theta_2} - \mathbf{J}_1^T\boldsymbol{F}_1 + \mathbf{J}_{m2}^T\boldsymbol{F}_{t_1} \\ (\ln(\mathbf{g}_{sc1}\,\mathbf{g}_{c1}^{-1}))^\vee \\ (\ln(\mathbf{g}_{sm2}\,(\mathbf{g}_{st1}\,\mathbf{T}_{12})^{-1}))^\vee \end{bmatrix} \tag{6}$$

In model $\boldsymbol{C}_\alpha$, the total number of equations and unknowns $\boldsymbol{x}_\alpha = [\boldsymbol{\theta_1}, \boldsymbol{\theta_2}, \boldsymbol{F}_{t_1}, l_1, f_1]^T$ is $n_1 + n_2 + 10$, thus the equilibrium configuration of the mechanism can be determined by solving the system of nonlinear equations. The system Jacobian matrix of this nonlinear algebraic equation system with respect to all unknown variables can be expressed as

$$\boldsymbol{\nabla}_{C_\alpha} = \begin{bmatrix} \mathbf{K}_{\theta_1} - \mathbf{K}_{\mathbf{J}_{t_1}} & \mathbf{0} & -\mathbf{J}_t^T & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{K}_{\theta_2} - \mathbf{K}_{\mathbf{J}_{m2}} - \mathbf{K}_{\mathbf{J}_1} & \mathbf{J}_m^T & \mathbf{J}_{\tau_2,l} & \mathbf{J}_{\tau_2,f} \\ \mathbf{0} & \mathbf{J}_1 & \mathbf{0} & \mathbf{J}_{1,l} & \mathbf{0} \\ -\mathbf{J}_{t_1} & \mathbf{J}_{m2} & \mathbf{0} & \mathbf{0} & \mathbf{0} \end{bmatrix} \tag{7}$$

The calculation method for each block in the matrix can refer to previous work. Subsequently, the system's equilibrium state can be solved through Newton-Raphson iteration, i.e.:

$$\boldsymbol{x}_{\alpha,k+1} = \boldsymbol{x}_{\alpha,k} - (\boldsymbol{\nabla}_{\boldsymbol{C}_\alpha})_k^{-1}\,\boldsymbol{C}_{\alpha,k} \tag{8}$$

Next, consider the deformation within plane β. Similar to the equivalent method in plane α, the kinematic-static model of the mechanism in plane β can be obtained as

$$\boldsymbol{C}_\beta(\boldsymbol{x}_\beta) = \begin{bmatrix} \boldsymbol{\tau_3} \\ \boldsymbol{\tau_4} \\ \boldsymbol{y}_{c2} \\ \boldsymbol{y}_{t2} \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\theta_3}\boldsymbol{\theta_3} - \mathbf{J}_{t_3}^T\boldsymbol{F}_{t_3} \\ \mathbf{K}_{\theta_4}\boldsymbol{\theta_4} - \mathbf{J}_2^T\boldsymbol{F}_2 + \mathbf{J}_{m4}^T\boldsymbol{F}_{t_3} \\ (\ln(\mathbf{g}_{sc2}\,\mathbf{g}_{c2}^{-1}))^\vee \\ (\ln(\mathbf{g}_{sm4}\,(\mathbf{g}_{st3}\,\mathbf{T}_{34})^{-1}))^\vee \end{bmatrix} \tag{9}$$

with $\boldsymbol{x}_\beta = [\boldsymbol{\theta_3}, \boldsymbol{\theta_4}, \boldsymbol{F}_{t_3}, l_2, f_2]^T$. Based on this model, the iterative solution of equilibrium configurations can be achieved by analytically computing the system Jacobian matrix similar to (7). When segmenting the elastic rod in plane β, the segment lengths from O_2 to the free end should match those in plane α to ensure that the equivalent joints are at the same positions, thereby allowing the superposition of bending effects from both planes.

The complete modeling process should follow the procedure described below. For a given target contact point position $\boldsymbol{p}_{ct}$, since the local coordinate frame $\{\mathrm{S}_\alpha\}$ of the mechanism in plane α coincides with the global coordinate frame $\{\mathrm{S}\}$, it can be projected onto plane α as the point $\boldsymbol{p}_{ct}^\alpha$:

$$\boldsymbol{p}_{ct}^\alpha = \boldsymbol{p}_{ct} - [\boldsymbol{p}_{ct}\cdot\boldsymbol{n}_\alpha]\boldsymbol{n}_\alpha \tag{10}$$

where $\boldsymbol{n}_\alpha = [0,1,0]^T$ is the normal vector of plane α. Taking $\boldsymbol{p}_{ct}^\alpha$ as the target contact point and solving through (6) and (7), the configuration of the mechanism in plane α can be obtained. Since the mechanisms in the two planes are connected in series, the local coordinate frame $\{\mathrm{S}_\beta\}$ of the mechanism in plane β can be derived from the local coordinate frame at the connection between the four-bar linkage and the whisker body. Specifically, the basis vectors of the local coordinate frame $\{\mathrm{S}_\beta\}$ expressed in the global coordinate frame $\{\mathrm{S}\}$ satisfy:

$$\begin{aligned} \boldsymbol{x}_\beta &= [0,1,0]^T \\ \boldsymbol{y}_\beta &= \boldsymbol{u}_{2n_m} \\ \boldsymbol{z}_\beta &= -\boldsymbol{u}_{1n_1+2} \end{aligned} \tag{11}$$

where $\boldsymbol{u}_{2n_m}$ denotes the u-axis direction of the local coordinate frame at the connection between the whisker shaft and the four-bar linkage in plane α. This direction is tangential to the whisker body and points toward the free end. $\boldsymbol{u}_{1n_1+2}$ denotes the u-axis direction of the local coordinate frame at the end of the first chain (i.e., the four-bar linkage) in plane α; this direction is parallel to the upper link and points toward the whisker shaft. Similarly, the target contact position can be projected onto plane β as the point $\boldsymbol{p}_{ct}^\beta$:

$$\boldsymbol{p}_{ct}^\beta = \boldsymbol{p}_{ct} - [(\boldsymbol{p}_{ct} - \boldsymbol{p}_{O_1})\cdot\boldsymbol{n}_\beta]\boldsymbol{n}_\beta \tag{12}$$

where the normal vector of plane β coincides with the z-axis direction $\boldsymbol{z}_\beta$ of frame $\{\mathrm{S}_\beta\}$.

After completing the calculations in planes α and β respectively, the deformations of the two elastic plates can be obtained, i.e., the first n_1 joint displacements of $\boldsymbol{\theta}_1$ and the first n_2 joint displacements of $\boldsymbol{\theta}_3$. Furthermore, the overall deformation of the whisker can be reconstructed in the global coordinate frame $\{\mathrm{S}\}$ as:

$$\begin{aligned} \mathbf{g}_{wt} = &\exp(\hat{\boldsymbol{\xi}}_{\alpha1}\theta_{\alpha1})\exp(\hat{\boldsymbol{\xi}}_{\beta1}\theta_{\beta1})\cdots \\ &\exp(\hat{\boldsymbol{\xi}}_{\alpha n_w}\theta_{\alpha n_w})\exp(\hat{\boldsymbol{\xi}}_{\beta n_w}\theta_{\beta n_w})\,\mathbf{g}_{wt,0} \end{aligned} \tag{13}$$

where $\mathbf{g}_{wt}$ is the end frame of the whisker body relative to $\{\mathrm{S}\}$, and the subscripts $\theta_{\alpha i}$ and $\theta_{\beta i}$ denote the joint displacements obtained from the calculations in plane α and β, respectively. For the segment OO_1 of the whisker body, $\theta_{\beta i} = 0$, and $\theta_{\alpha i}$ takes the deformation of the corresponding

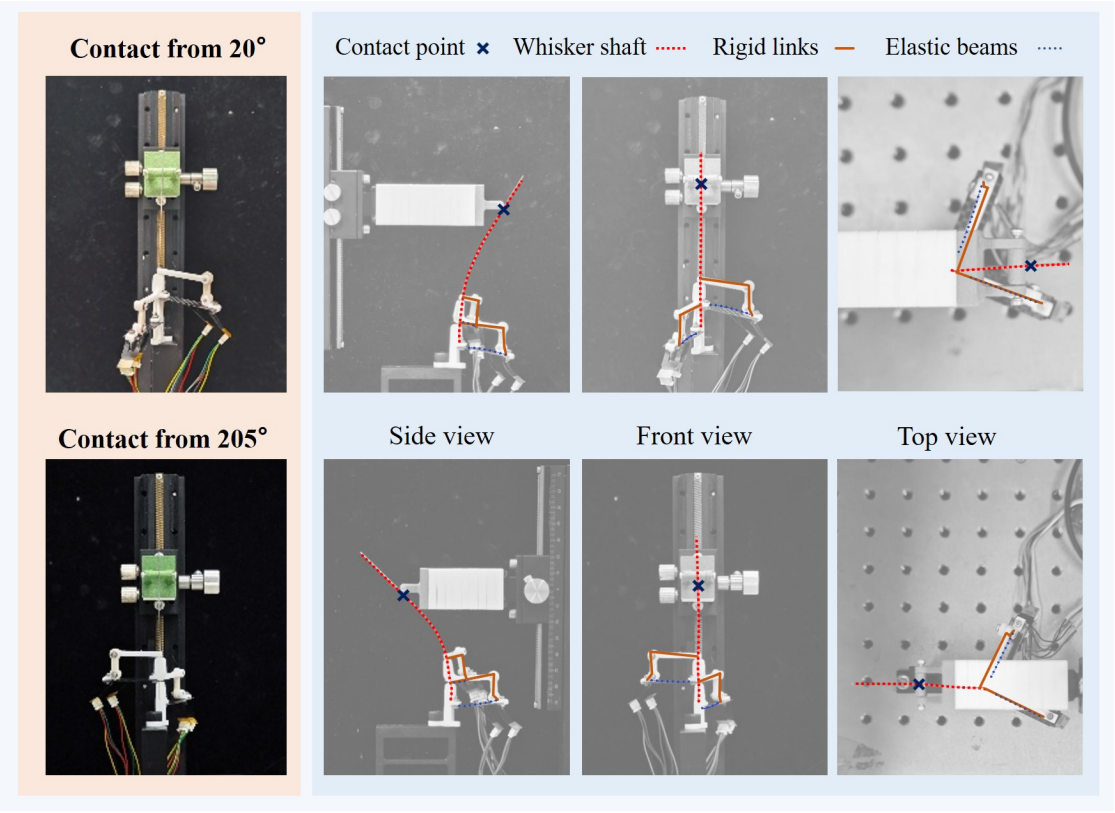


Figure 4: Comparison Results Between Model Calculation and Photographs of Whisker Sensor States Under Spatial Contact From Different Directions.

joint in θ_2. For the segment O_1O_2, $\theta_{\alpha i} = 0$, and $\theta_{\beta i}$ takes the deformation of the corresponding joint in θ_4. For the segment from O_2 to the free end of the whisker shaft, $\theta_{\alpha i}$ and $\theta_{\beta i}$ take the deformations of the corresponding joints in θ_2 and θ_4, respectively.

Figure 4 presents a comparison between the actual photographs and the whisker states obtained from the above modeling under two different spatial contact conditions, captured from three distinct viewing angles. The comparison demonstrates that the contact point positions and the states of each link obtained through modeling calculations are generally consistent with the actual conditions. The discrepancies between them mainly arise from non-modeled factors such as clearances existing during the assembly process of the whisker sensor, and frictional forces between the internal links of the sensor and the inner wall. Comparisons between the model's computational results and photos under more configurations are presented in the supplementary materials. The calibration process, convergence behavior, and sensitivity to parameters are also included in the supplementary materials.

4. Multimodal Sensing Methods

Based on the sensor architecture and sampling system proposed in the previous section, this section elucidates the specific methodologies employed by the whisker sensor to achieve multimodal sensing, encompassing both direct contact localization and airflow characterization.

4.1. Direct Contact Sensing

To characterize the spatial position of external contact relative to the whisker sensor, a cylindrical coordinate system was established, with its axes coinciding with the spatial coordinate system {S}, as shown in Figure 5.The position of the contact point can be represented by three coordinates ϕ_c, r_c, and z_c. These coordinates satisfy the following transformation relationship with the spatial Cartesian coordinate system {S}:

$$\begin{aligned} x_{c,S} &= r_c cos(\phi_c) \\ y_{c,S} &= r_c sin(\phi_c) \\ z_{c,S} &= z_c \end{aligned} \tag{14}$$

Since the local deformation at the strain gauge attachment position can be solved through the process described in the previous section under known external contact constraints, conversely, external contact sensing can also be achieved by introducing local deformation constraints to replace the geometric constraints of external contact. However, the proposed whisker sensor is inevitably affected by factors such as shaft clearance, printing accuracy, and internal friction, which are difficult to fully consider in modeling. Therefore, this paper proposes a cascaded residual learning architecture that combines the kinetostatics model with a data-driven neural network. This architecture uses the

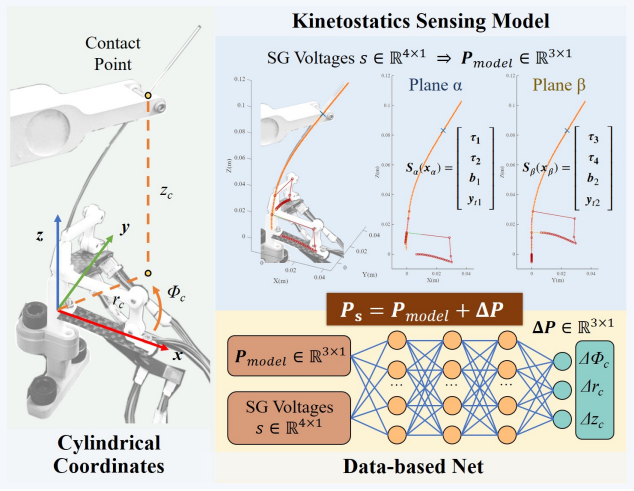


Figure 5: Direct Contact Sensing of the Poposed Whisker Sensors

theoretical model to provide physically consistent baseline predictions while employing the neural network to learn the systematic deviations caused by unmodeled factors, thereby improving prediction accuracy while ensuring generalization capability.

Let the measured voltage values of the four strain gauges of the whisker sensor be $\mathbf{s} = [s_{11}, s_{12}, s_{21}, s_{22}]^{\mathrm{T}} \in \mathbb{R}^{4\times1}$, and the corresponding spatial contact point coordinates be $\mathbf{p} = [\phi_c, r_c, z_c]^{\mathrm{T}} \in \mathbb{R}^3$. First, the kinetostatics model is used to solve the inverse problem and obtain the theoretical predicted coordinates $\mathbf{p}_{\text{model}}$. This theoretical prediction has high accuracy under ideal conditions, but due to unmodeled factors, there exists a systematic deviation $\mathbf{e} = \mathbf{p}_{\text{true}} - \mathbf{p}_{\text{model}}$ in the prediction results.

To correct this deviation, a neural network $\mathcal{N}$ is introduced, which takes the concatenated vector of the theoretical prediction $\mathbf{p}_{\text{model}}$ and the original strain readings $\mathbf{s}$ as input to learn the residual $\Delta\mathbf{p}$. The final contact point coordinates are obtained by summing the theoretical prediction and the residual:

$$\mathbf{p}_{\text{s}} = \mathbf{p}_{\text{model}} + \Delta\mathbf{p} \tag{15}$$

4.1.1. Kinetostatics-Based Sensing Model

Similar to the kinetostatic modeling with known external contact conditions, the sensing process is also carried out in the two mutually perpendicular planes described above, and the coordinates of the contact point are then obtained by superposition. First, consider the sensing process in plane α, the physical contact constraints in the equilibrium equations (6) are replaced by the joint displacement constraints associated with the strain gauges, yielding the sensing model:

$$\boldsymbol{S}_\alpha(\boldsymbol{x}_\alpha) = \begin{bmatrix} \boldsymbol{\tau_1} \\ \boldsymbol{\tau_2} \\ \boldsymbol{b}_1 \\ \boldsymbol{y}_{t1} \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\boldsymbol{\theta_1}}\,\boldsymbol{\theta_1} - \mathbf{J}_{t_1}^T \boldsymbol{F}_{t_1} \\ \mathbf{K}_{\boldsymbol{\theta_2}}\,\boldsymbol{\theta_2} - \mathbf{J}_1^T \boldsymbol{F}_1 + \mathbf{J}_{m2}^T \boldsymbol{F}_{t_1} \\ \mathbf{S}_{e1}\boldsymbol{\theta}_1 - \boldsymbol{s}_1 \\ (\ln(\mathbf{g}_{sm2}\,(\mathbf{g}_{st1}\,\mathbf{T}_{12})^{-1}))^\vee \end{bmatrix} \tag{16}$$

where $\boldsymbol{b}_1 = \mathbf{S}_{e1}\boldsymbol{\theta}_1 - \boldsymbol{s}_1$ is the $m_1 \times 1$ constraint introduced by the strain gauges, assuming that a number of m_1 strain gauges are attached. Here, $\mathbf{S}_{e1}$ is a constant selection matrix indicating the position of each strain gauge such that

$$\mathbf{S}_{e1} = \left[\mathbf{S}_{11}, \cdots, \mathbf{S}_{1j}, \cdots, \mathbf{S}_{1m}\right]^T \in \mathbb{R}^{m_1\times(n_1+2)} \tag{17}$$

where $\mathbf{S}_{1j} = [0, \cdots, 0, 1, \cdots, 1, 0, \cdots, 0]$ with its $n_{1j,p}$ to $n_{1j,q}$ elements being 1 and other elements being 0. $n_{1j,p}$ and $n_{1j,q}$ indicate the first and last joint covered by each strain gauge.

The sensing model $\boldsymbol{S}_\alpha$ consists of $n_1 + n_2 + 8 + m_1$ equations and $n_1 + n_2 + 10$ unknown variables. Therefore, when two strain gauges are attached as the proposed design, the model can be solved using the iterative method. Since all variables and most equations in (16) are consistent with those in (6), the system Jacobian matrix can be obtained in a similar manner, and the equilibrium state can be solved using the Newton-Raphson method. Similarly, the sensing model in the β plane can be expressed as:

$$\boldsymbol{S}_\beta(\boldsymbol{x}_\beta) = \begin{bmatrix} \boldsymbol{\tau_3} \\ \boldsymbol{\tau_4} \\ \boldsymbol{b}_2 \\ \boldsymbol{y}_{t2} \end{bmatrix} = \begin{bmatrix} \mathbf{K}_{\boldsymbol{\theta_3}}\,\boldsymbol{\theta_3} - \mathbf{J}_{t_3}^T \boldsymbol{F}_{t_3} \\ \mathbf{K}_{\boldsymbol{\theta_4}}\,\boldsymbol{\theta_4} - \mathbf{J}_2^T \boldsymbol{F}_2 + \mathbf{J}_{m4}^T \boldsymbol{F}_{t_3} \\ \mathbf{S}_e\boldsymbol{\theta}_4 - \boldsymbol{s}_2 \\ (\ln(\mathbf{g}_{sm4}\,(\mathbf{g}_{st3}\,\mathbf{T}_{34})^{-1}))^\vee \end{bmatrix} \tag{18}$$

This model also requires at least two strain gauges for sensing calculations. Through the two in-plane sensing models (16) and (18), two sets of deformations $\boldsymbol{\theta}_2$ and $\boldsymbol{\theta}_4$ of the whisker's equivalent joints can be obtained, as well as the curved distances l_1 and l_2 from the external contact point to their respective origins along the whisker shaft in the two local coordinate frames $\{\mathrm{S}_\alpha\}$ and $\{\mathrm{S}_\beta\}$. With the known lengths of each equivalent segment, l_1 and l_2 can be converted into equivalent segment numbers n_{c1} and n_{c2}, respectively. When the contact point is fixed, its equivalent segments in the two planes should be the same. However, in practical sensing calculations, due to measurement and modeling errors, $n_c = (n_{c1}+n_{c2})/2$ is taken as the equivalent segment number where the external contact occurs. The coordinates of the external contact point can then be calculated using the following equation:

$$\begin{aligned} \mathbf{g}_c = &\exp(\hat{\boldsymbol{\xi}}_{\alpha1}\theta_{\alpha1})\exp(\hat{\boldsymbol{\xi}}_{\beta1}\theta_{\beta1})\cdots \\ &\exp(\hat{\boldsymbol{\xi}}_{\alpha n_c}\theta_{\alpha n_c})\exp(\hat{\boldsymbol{\xi}}_{\beta n_c}\theta_{\beta n_c})\,\mathbf{g}_{c,0} \end{aligned} \tag{19}$$

Using the transformation relationship described in (14), the contact point coordinates can be unified into the cylindrical coordinate system.

4.1.2. Contact Data Acquisition and Validation

To collect training and validation data, three coordinates were discretized within the spatial domain. ϕ_c was discretized from 0 to 2π with an interval of $\pi/18$, r_c was discretized from 5 *mm* to 25 *mm* with an interval of 5 *mm*, and z_c was discretized from 50 *mm* to 95 *mm* with an interval of 15 *mm*, generating a total of 720 contact points. This discretization method covers more than half of the region

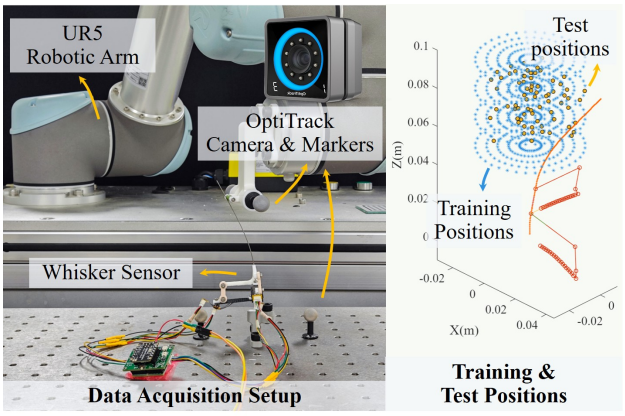


Figure 6: Experimental Setup for Data Acquisition with Contact Positions for Training and Test.

above the point O_2 of the whisker shaft. Due to the outward expansion of the follicle area, structural constraints make it difficult to generate external contacts near the whisker base in practical applications. Therefore, the selected data acquisition region already encompasses the common contact range, and this discretization method is considered suitable for most single-point contact scenarios.

As shown in Figure 6, to apply external contacts, a UR5 robotic arm was controlled to bring an object into contact with the whisker sensor while the strain gauge readings were recorded. Simultaneously, an OptiTrack motion capture system was used to measure the precise contact point positions under each planned contact configuration. The data collection process was repeated three times. At the same contact position, the maximum voltage deviation of the strain gauges across the three trials was below $2\,mV$. Considering that the maximum voltage variation of the strain gauges exceeds $100\,mV$, the developed sensor exhibits good repeatability. In addition to the 720 contact points, another 75 sets of data with random coordinates in the discretization range were also collected to validate the contact sensing performance. These datasets were collected independently and were not used in any stage of training.

For each training sample, the theoretical model was first used to compute $\mathbf{p}_{\text{model},i} = \mathcal{M}_{\text{inv}}(\mathbf{s}_i)$ offline, and the true residual $\Delta\mathbf{p}_{\text{true},i} = \mathbf{p}_{\text{true},i} - \mathbf{p}_{\text{model},i}$ was obtained. Subsequently, a backpropagation (BP) neural network with seven inputs (three-dimensional $\mathbf{p}_{\text{model}}$ and four-dimensional $\mathbf{s}$) and three outputs was used to establish the mapping between the strain gauge voltage readings and the residuals of the three spatial coordinates. The network comprises three hidden layers with 5, 10, and 10 neurons, respectively. The hidden layers use a hyperbolic tangent sigmoid transfer function, while the output layer uses a linear transfer function. Training is performed using the Bayesian regularization backpropagation algorithm.

Using the cascaded residual learning architecture, the three coordinates of the contact point can be predicted through the trained network. Figure 7 presents the network

Table 1
Contact Sensing Error Using Different Models

Model	Mean Error (Within-Range)		
	ϕ_c (rad)	r_c (mm)	z_c (mm)
Physics-only	0.09	4.74	8.59
Data-driven only	0.08	3.61	7.40
Hybrid	**0.06**	**2.94**	**5.95**
Model	**Mean Error (Expand)**		
	ϕ_c (rad)	r_c (mm)	z_c (mm)
Physics-only	0.08	4.88	8.36
Data-driven only	0.14	5.58	10.38
Hybrid	**0.08**	**3.30**	**7.35**

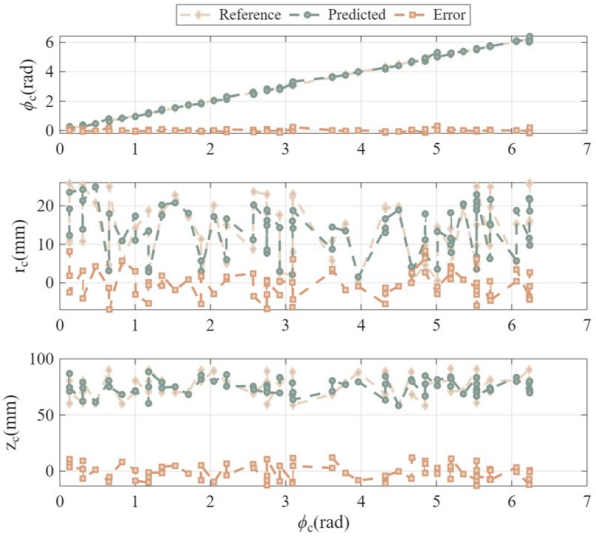


Figure 7: Testing Result of Contact Sensing on Different Directions.

prediction results of the 75 test data sets on different directions ϕ_c. The average sensing errors for ϕ_c, r_c, and z_c are $0.06\,rad$, $2.94\,mm$, and $5.95\,mm$, respectively. The distribution of error shows that the contact sensing results are quite consistent across different angles, which indicates that the developed whisker sensor exhibits comparable performance across different contact positions. For comparison, contact sensing was performed on the same test data using the purely physical model and the purely data-driven network, i.e., with $\mathbf{s}$ as the input and $\mathbf{p}$ as the output. The resulting mean sensing errors for the three methods are shown in Table 1. To verify the generalization of the perception method, another 30 sets of random data that were slightly out of the training range were collected, with r_c in the range of $25 - 30\,mm$, and z_c in the range of $45 - 50\,mm$ and $95 - 100\,mm$. The sensing results for the three models on this dataset is also shown in Table 1.

The results demonstrate that the proposed hybrid model consistently outperforms both the physics-only and purely data-driven approaches across both datasets. On the within-range test points, the hybrid model reduces the mean error by around 30% compared to the physics-only model, and by around 20% compared to the purely data-driven model. On

the expanded-range test points, the hybrid model maintains reasonable accuracy, confirming its enhanced generalization capability. In contrast, the purely data-driven network shows comparable performance to the hybrid model on the within-range data but experiences a larger drop in accuracy on the expanded-range data, while the physic-only model maintains a stable performance, indicating that the physics-based prior provides better extrapolation capability.

4.2. Airflow Sensing

In addition to detecting direct contact, the developed sensor is capable of perceiving airflow characteristics. In this section, the sensor's abilities to determine the direction of a wind source and measure flow speed are validated.

For airflow sensing, a fin-like structure is attached at the free end of the whisker sensor. The fin shape is designed based on the following quantitative considerations. The whisker shaft has a projected area of approximately $48\,mm^2$. To justify a concentrated-force approximation, the fin area should be at least 10 times the shaft area, requiring $A_{\min} \approx 480\,mm^2$. To avoid interference with adjacent whiskers, the dimension perpendicular to the shaft is limited to approximately $40\,mm$. Within this constraint, a cross-section of around $30\,mm \times 30\,mm$ was selected, providing an area of $900\,mm^2$ while ensuring compactness and ease of 3D printing fabrication. Similar fin-based designs for airflow enhancement have been reported in prior work [16, 40].

4.2.1. Wind Source Direction Identification

Similar to direct contact, when wind blows toward the whisker from different directions, the two elastic beams undergo varying degrees of deformation. Therefore, by analyzing the magnitude of the output voltages from the strain gauges attached to each beam, the direction of the wind source relative to the whisker sensor can be determined. However, unlike direct contact, airflow exerts a distributed pressure over the whisker shaft and the attached fin structure. To obtain a tractable relationship between strain gauge signals and airflow parameters, a simplified quasi-static model is adopted under the condition of stable flow. Assuming that the wind-induced force is dominated by the pressure on the fin and the resultant aerodynamic force can be treated as a concentrated load at the center of the fin (15 mm from the free end of the shaft). Similar assumption methods are used in research related to sensor modeling [9, 41].

Under this assumption, the kinetostatics model established in Section 3 was used to compute the deformation of the whisker sensor by applying concentrated forces at the center of the fin-like structure on the whisker shaft from various directions ϕ. In this way, the strain gauge responses corresponding to different wind source directions can be simulated. In the theoretical computation, 72 loading directions uniformly distributed at 5° intervals were selected, and for each direction three different force magnitudes were applied. Fig. 8 shows the variation of the ratio k_U — defined as the ratio of the readings from the strain gauges on the two elastic beams that exhibit the largest deformation — with respect to the loading direction obtained from the model. The

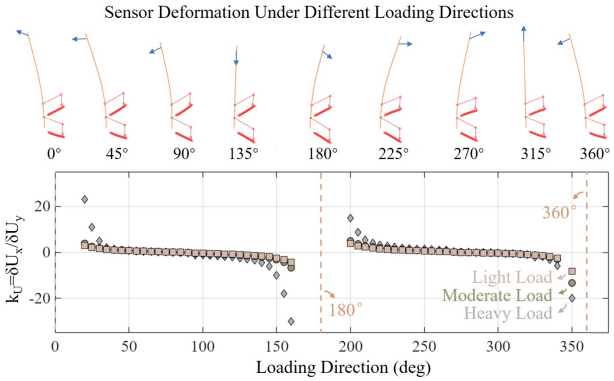


Figure 8: Relationship Between Wind Source Direction and Strain Gauge Deformation Calculated Through Theoretical Model.

Table 2
Sign of δU_x and δU_y under different wind direction

Wind Direction (°)	0-90	90-180	180-270	270-360
Sign of δU_x	+	-	-	+
Sign of δU_y	+	+	-	-

results indicate that near 0° or 180°, the ratio becomes very large because the strain gauge in the y-direction undergoes only small deformation. Moreover, the curves for different force magnitudes share a similar shape but differ in steepness. Consequently, normalization of the data is necessary. This theoretical analysis provides a basis for the subsequent fitting of the experimental data.

As shown in Fig. 9, to validate the direction identification ability, a whisker sensor was mounted on a rotational platform, allowing the wind source to blow toward the whiskers from any direction. A propeller was positioned 175 mm away from the center of the whisker as the wind source, with its central axis aligned with the center of the whisker's fin-like structure, thereby ensuring that the whisker senses the airflow along the propeller's central axis. The rotational platform was rotated to 18 distinct orientations ϕ_w at 20° intervals. On each direction, the propeller was adjusted to 3 different rotational speeds. For each configuration, readings of the strain gauges are recorded.

Under the effect of wind, the strain gauge voltage $U(t)$ oscillates around a specific value. By analyzing a 4-second window of the oscillatory segment, the mean change δU relative to the initial value can be obtained. For the recorded data, the changes in voltage readings, denoted as δU_x and δU_y, from the strain gauges exhibiting the largest deformation magnitude on each of the two elastic beams were calculated.

The following procedure is followed to determine the direction of wind source. Firstly, the approximate directional range of the wind source can be estimated by examining the signs of δU_x and δU_y, with specific results presented in Table 2.

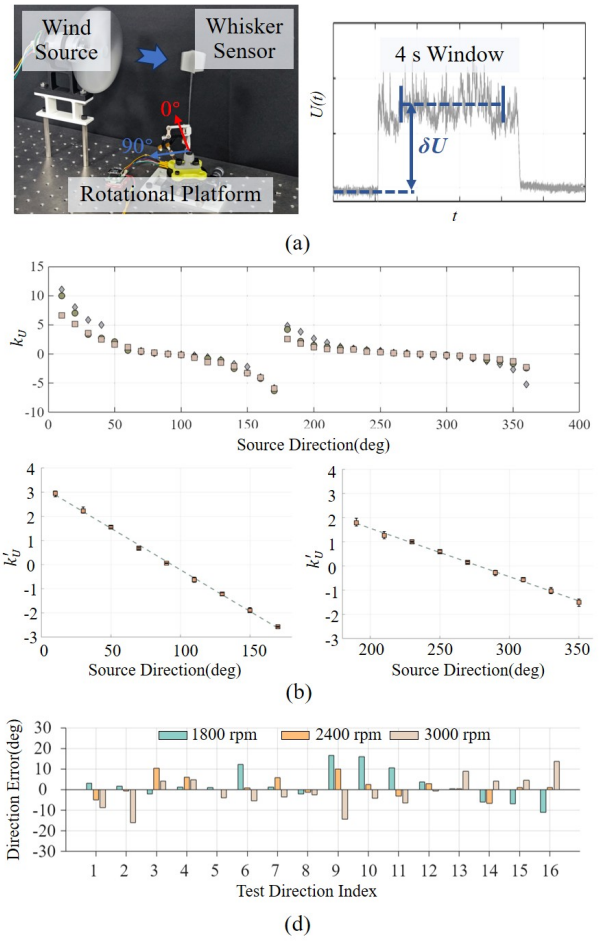


Figure 9: Wind Source Direction Identification Using the Proposed Whisker Sensor: (a) Experiment Setup with a Voltage Reading Example. (b) Result of k_U and k'_U Against Different Source Directions. (c) Identification Error on Different Directions and Rotational Speeds.

When wind blows from different directions, the ratio of the voltage changes from the two strain gauges, defined as $k_U = \delta U_x / \delta U_y$, varies. Fig. 9 shows the k_U values under different wind source directions, and these results are in close agreement with those obtained from the theoretical model. For directions of 0° or 180°, the value of δU_y is relatively small, resulting in a large value for k_U. Additionally, as indicated in Table 2, the sign of k_U undergoes an abrupt change around these two directions, leading to discontinuity in the value of k_U. Therefore, separate $\phi_w - k_U$ fitting procedures are applied for the two intervals corresponding to wind directions of $0-180°$ and $180-360°$, respectively. Since the k_U ratio is also influenced by wind speed, it must first be normalized through scaling:

$$k'_U = \text{sign}(k_U)\ln(1 + |k_U|/\beta) \tag{20}$$

where $\beta = 0.5$ is a constant for all the propeller rotational speeds. After scaling, under the same wind direction, the value of k'_U does not exhibit significant variation with changes in wind speed magnitude. Through linear fitting, the relationship between ϕ_w and k'_U is established separately for the two intervals, as shown in Fig. 9, where the coefficient of determination R^2 for the fitting results exceeds 0.98.

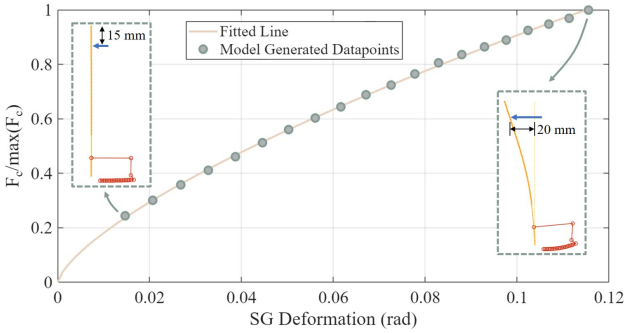


Figure 10: The Relationship Between the Magnitude of External Force and the Deformation of Strain Gauges Calculated by the Theoretical Model.

To validate the flow direction sensing capability of the developed sensor, the whisker sensor was rotated to 16 directions different from those in the fitting process. At each direction, the propeller was adjusted to three different rotational speeds. For these 48 configurations, the aforementioned procedure was followed. Firstly, the direction range was determined based on the signs of δU_x and δU_y. The value of k_U was then calculated and normalized to k'_U. And finally, it was substituted into the corresponding fitted curve to compute the direction ϕ_w. The direction identification error of the test configurations are presented in Fig. 9. The average directional sensing error across the 48 datasets was 5.41°.

4.2.2. Wind Speed Sensing

Based on the method described in the previous subsection, once the direction of the wind source is obtained, the x-axis of the whisker can be aligned with the wind source. In this orientation, the airflow can induce significant deformation of the whisker.

Similar to the previous section, the kinetostatics model is first used to compute the deformation of the strain gauges by applying loads of various magnitudes at the center of the fin-like structure in the x-direction, in order to obtain the relationship between the strain gauge response and the external force. Fig. 10 shows the theoretically calculated deformation curves of the strain gauge close to the whisker body under different external forces. After fitting, the relation satisfies:

$$F_w \propto dU^{0.70} \tag{21}$$

where F_w denotes the resultant aerodynamic force acting on the fin. For a bluff body immersed in a cross-flow, the aerodynamic drag force follows the well-established quadratic relationship with the incoming flow speed as

$$F_w = \frac{1}{2} C_D \rho A v_w^2 \tag{22}$$

where C_D is the drag coefficient of the fin, ρ is the air density, A is the projected area of the fin normal to the flow, and

v_w is the wind speed. This quadratic drag–velocity law has been extensively applied in whisker-inspired and cantilever-based flow sensors to relate the fluid dynamic loading to the structural response [42, 43]. Combining the mechanical transfer characteristic with the aerodynamic drag law yields the theoretical scaling relationship between the wind speed and the strain gauge output:

$$v_w \propto dU^{0.35} \tag{23}$$

To experimentally calibrate the actual $v_w - dU$ relationship, the sensor was placed 125 mm away from the propeller, with its x-axis parallel to the propeller center axis. The propeller was adjusted to 17 distinct rotational speeds ranging from 1080 rpm to 3000 rpm. At each speed setting, the voltage readings from the strain gauges were recorded, while the actual wind speed at the sensor's location was simultaneously measured using an anemometer (DLX-ANE2302, accuracy: 0.1 m/s). Here, only the reading from the strain gauge oriented along the propeller axis, which exhibits larger amplitude variations, was utilized. Similar to the previous cases, its mean change δU relative to the initial value was calculated for each configuration.

Following the theoretical analysis, a power-law curve passing through the origin was used to fit the relationship between voltage and wind speed. The relationship satisfies:

$$v_w = 2.68\delta U^{0.27} \tag{24}$$

The fitted result yields an R^2 value of 0.98, accurately reflecting the $v_w - \delta U$ relationship. From the fitted equation, the exponent 0.27 is lower than the theoretical prediction 0.35. This discrepancy can be attributed to several factors not captured by the simplified theoretical model. First, the drag coefficient of the flat-plate fin is not strictly constant but decreases with increasing Reynolds number within the operating range, causing the actual aerodynamic force to grow more slowly than v_w^2 . Second, the kinetostatics model assumes ideal frictionless joints and perfectly decoupled planar bending, whereas the physical prototype exhibits minor mechanical losses and cross-plane coupling that can attenuate the strain response at larger deflections. Despite these deviations, the close agreement between the overall trend of the theoretical prediction and the experimental calibration confirms that the power-law form is physically well motivated, and the empirical fit provides an accurate mapping for the intended operating conditions.

To validate the accuracy of wind speed sensing, the propeller was adjusted to four different rotational speeds: 1200, 1800, 2400, and 3000 rpm. The developed sensor was positioned along the central axis at five distances from the propeller: 100, 150, 200, 250, and 300 mm, for wind speed measurement. The results of the wind speed measurements are shown in Fig. 11. The dashed lines represent the reference wind speed–distance relationships measured by the anemometer at different rotational speeds, while the scattered points of different shapes indicate the wind speeds perceived by the whisker sensor at the respective rotational

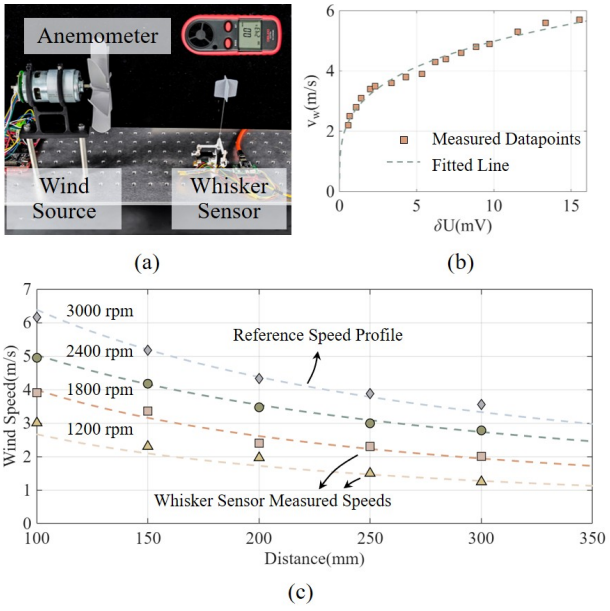


Figure 11: Wind Speed Sensing Using the Proposed Whisker Sensor: (a) Experiment Setup. (b) Fitted Relationship Between Sensor Voltage and Wind Speed. (c) Wind Speed Sensing Result for the Propeller.

speeds. Based on the sensor's noise characteristics, mechanical limits, and the fitted calibration curve, the theoretical detectable wind speed is approximately $1.0-7.7\,m/s$ using a 3 times noise criterion. Due to the limited output range of the propeller, the wind speeds tested ranged from approximately 1.2 to $6.2\,m/s$, which falls entirely within the theoretical measurement range. The average wind speed measurement error in this range is 0.12 m/s, and its variation at different distances aligns with the measured propeller wind speed curves, which demonstrates that the developed sensor can accurately sense wind speed and can be used to reliably reconstruct the wind speed–distance relationship within a airflow field.

5. Demonstration on Mobile Robots

Animals such as rats exhibit anemotaxis—the ability to sense airflow using their whiskers and move toward the wind source—while also utilizing whiskers to detect direct contact and navigate around obstacles in their environment. Biologically, this multimodal capability relies on functional specialization: supra-orbital whiskers are dedicated to airflow sensing, whereas mystacial whiskers primarily mediate tactile exploration. Inspired by this strategy, we integrate multiple copies of the same whisker sensor into a mobile robot, which are assigned to a distinct modality. This approach preserves the multimodal perception capability observed in biology while avoiding the need for sensor redesign—reducing fabrication complexity, lowering cost, and simplifying real-time signal interpretation.

As shown in Fig. 12, a four-wheeled mobile robot was developed, where three whisker sensors are attached. The

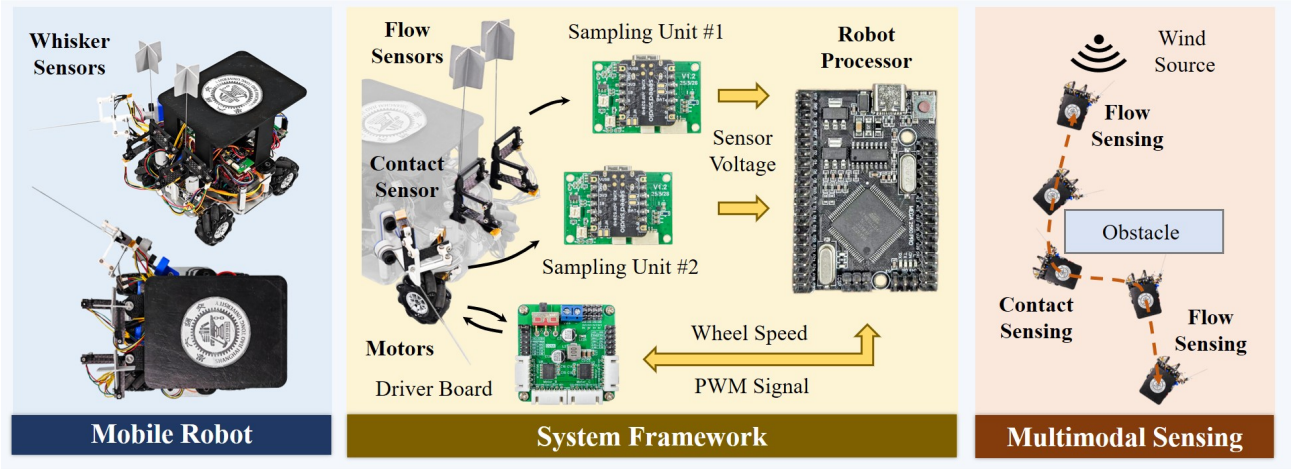


Figure 12: Environmental Perception Applications of the Proposed Sensors on a Mobile Robot.

two sensors mounted on the upper section of the mobile robot are designated for fluid sensing, while the sensor on its side is employed for direct contact sensing. To acquire strain gauge voltages, the mobile robot is equipped with two sets of sampling circuit units, one set samples the four strain gauges required for direct contact sensing, while the other samples the strain gauges on each of the two airflow-sensing whiskers. Through serial communication, the whisker sampling units' microcontroller transmit the acquired voltage data to the robot's main processor (Arduino Mega 2560), where the data is processed, and robot control is subsequently executed based on the sensor information. Given the utilization of fitted networks and models, the main processor of the mobile robot is capable of efficiently processing the strain gauge voltage information and converting it into contact points or desired fluid characteristics. The robot completes the entire sensing-computation-control loop at a frequency exceeding 30 Hz. Additionally, the entire robotic system is powered by a single 12V lithium battery, eliminating the need for external cables and enabling wireless environmental exploration with low power consumption.

Emulating the anemotactic behavior of creatures, two scenarios are presented based on the mobile robot. As shown in Fig. 13, in the first scenario, the mobile robot was placed in an open area and remained stationary. A handheld fan was used to direct airflow toward the whisker sensors, with the fan's central axis parallel to the ground and aimed at the fin-like structure at the free end of the whiskers. Through the computational procedure described in Section 4.2.1, the two whisker sensors could independently determine the direction of the fan relative to each sensor. Given that the positions of the two sensors relative to the center of the mobile robot are known, the direction of the fan relative to the robot's center could be calculated. Subsequently, the robot rotated in place to align itself with the direction of the fan and then proceeded to move toward it. To test the effectiveness of the anemotactic motion, four different wind source directions were randomly selected around the robot, and three repeated

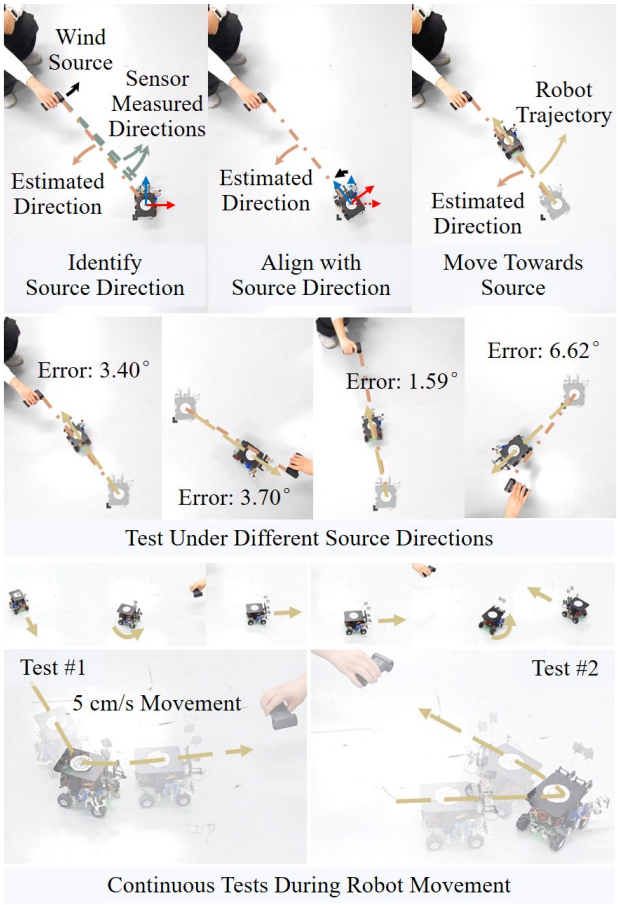


Figure 13: Anemotactic Behavior of the Mobile Robot Based on the Proposed Whisker Sensors.

experiments were conducted for each wind direction. Experimental results demonstrated that, when airflow was directed from multiple different orientations, the mobile robot could reliably move toward the fan based on the direction estimated

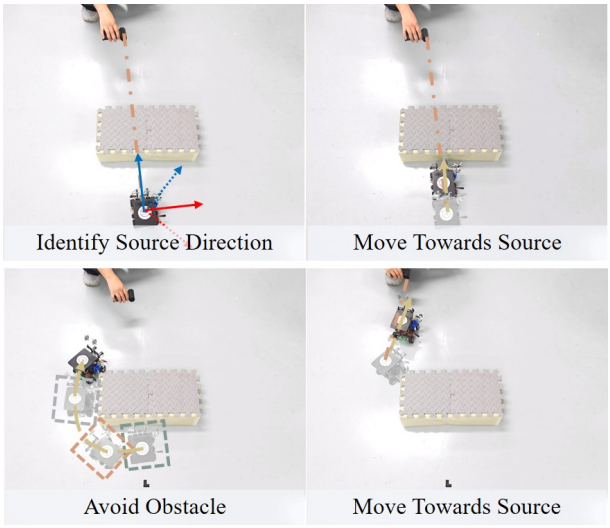


Figure 14: Demonstration of Multimodal Sensing of Obstacle and Airflow in Unstructured Environment on Mobile Robot.

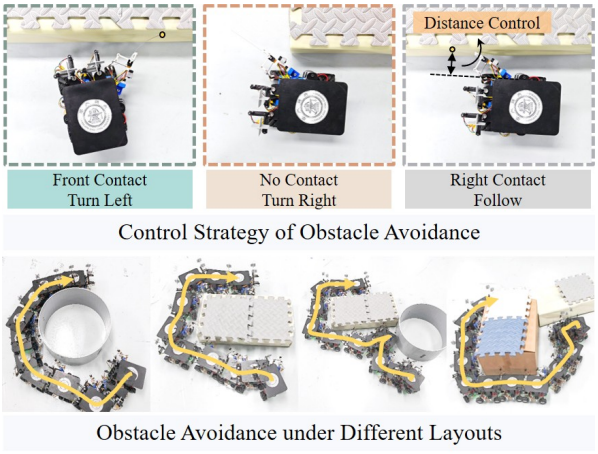


Figure 15: Obstacle Avoidance Strategy and Test for Four Different Obstacle Layouts.

from the developed sensors. For the tested wind source directions, the average error in the approaching direction was 3.83°.

In addition to the stationary sensing described above, the whisker sensors can perceive wind direction during robot motion at low speeds, where the external wind signal can be distinguished from the movement-induced noise. To demonstrate this, additional experiments are conducted in which the robot moved forward at a speed of approximately $5\,cm/s$, while a handheld fan directed airflow toward the whisker sensors. The robot successfully turned and moved toward the fan in response to the perceived wind direction. This process was repeated multiple times under different airflow directions to confirm repeatability. Two representative trials with different fan directions are shown in Fig. 13. The complete experimental process is provided in the supplementary video [1].

Beyond anemotaxis, the multimodal sensing capabilities of the developed sensor also enable the robot to possess multiple sensing functions. In the second scenario, a wall is placed between the fan and the robot, allowing airflow to pass through the obstacle while blocking the robot's direct path, as shown in Fig. 14. Initially, as the fan directs airflow toward the sensors, the mobile robot turns and moves toward the fan. When the side whisker sensor come into direct contact with the obstacle, the robot switches to an obstacle avoidance mode. After navigating around the obstacle, the robot re-detects the direction of the fan relative to itself and resumes movement toward it.

Specifically, for obstacle avoidance mode, the robot follows a right-side wall-following strategy [44]: upon detecting an obstacle on its right side, it adjusts the rotational speeds of its left and right wheels based on the contact point location, maintaining a relatively constant distance from the obstacle; if an obstacle is detected ahead, the robot executes a left turn; and when no obstacles are detected either ahead or to the right, it performs a right turn. To evaluate the obstacle avoidance capability, the robot was tested under four different obstacle layouts, cylindrical object, rectangular object, combination of cylindrical and rectangle objects, and combination of rectangular objects. For each layout, obstacle avoidance was repeated three times. In all trials, the robot successfully bypassed the obstacles. For cylindrical and rectangular obstacles, the navigation time is about $20\,s$, and for the combined obstacles, the navigation time is about $30\,s$[2]. Figure 15 shows representative trajectories of the robot for the four layouts, illustrating the robot's ability to adapt its path to different obstacle.

Experiments demonstrate that the robot can successfully bypass obstacles and eventually reach the location of the fan[3]. This verifies that the developed whisker sensor can achieve multimodal sensing on a robotic platform, equipping the robot with the ability to traverse complex, unstructured environments.

6. Discussion

The proposed whisker sensor offers a compact, low-cost solution that unifies contact localization and airflow perception within a single hardware platform, using only four strain gauges. And the successful integration on a mobile robot demonstrates the practical viability and real-time capability of the system. The performance is achieved under well-defined assumptions that should be considered when extending the sensor to more general conditions.

For contact sensing, the two perpendicular four-bar mechanisms, guided by baffle plates, largely decouple bending into two planes, which, together with the slender-shaft assumption and single-point contact, makes the contact coordinates observable from four strain readings. Violations (e.g., multi-point contacts or large axial forces) would weaken the unique mapping. The axial coordinate shows

[1] See supplementary materials - S1.

[2] See supplementary materials - S2.

[3] See supplementary materials - S3.

a relatively larger error, which is consistent with the lower sensitivity of base moments to contact height. For obstacle avoidance, however, this is acceptable as navigation relies more on proximity and direction.

The airflow experiments were currently conducted under controlled, single-source, steady-flow conditions to establish a baseline relationship between strain gauge responses and airflow parameters. Turbulent or multi-source flows may degrade direction and speed estimation. Several preliminary experiments are conducted as detailed in the supplementary material. These include multi-source scenarios using a two-sensor array, investigation of the minimum averaging window and effect of both contact and airflow. A more systematic investigation of these aspects will be pursued in future work.

Despite these limitations, the sensor's mechanical simplicity, dual-modal capability, and model-informed learning strategy provide a solid basis for miniaturization and deployment in mobile robots.

7. Conclusion and Future Work

This paper presents a biomimetic whisker sensor for multimodal sensing of direct contact and airflow, which can be applied in environmental exploration of robotic systems. The proposed sensor is inspired by biological whiskers and features a simple, mechanically compliant structure. The proposed sensor features a simple, mechanically compliant architecture and leverages industrial-grade strain gauges on a four-bar mechanism with slender elastic beams. Using a model-based approach, the sensor localizes intermediate contact points along the whisker shaft and estimates their spatial coordinates with high reproducibility. Additionally, the sensor demonstrates the ability to perceive airflow characteristics, including wind direction and speed, by analyzing strain gauge responses under fluid-induced deformation. Experimental validations confirm the sensor's effectiveness in both contact and flow sensing, with mean errors within practical limits for robotic applications. Furthermore, implementation on a mobile robot showcases its capability of anemotactic behavior and obstacle avoidance in unstructured environments, confirming the potential for multimodal environmental perception. Future work will address the limitations identified above through several directions. For airflow perception, multi-source identification using multi-whisker arrays will be investigated. For contact perception, improvements will be conducted to handle multi-point contacts and simultaneous airflow-contact coupling. Additionally, fluid dynamics analysis of the fin geometry, miniaturization of the sensing system, and tighter integration of sensing elements will be pursued to improve spatial resolution and enable denser whisker arrays.

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